# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 02

**Full Syllabus (Chapters 1–14)** | Class 8 Maths | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** even | even × even = even |
| 2 | **(a)** 144 | Sum of first $n$ odd numbers $= n^2 = 144$ |
| 3 | **(a)** $6.3 \times 10^6$ | One non-zero digit before the decimal point |
| 4 | **(d)** D | I=1, V=5, X=10, L=50, C=100, D=500, M=1000 |
| 5 | **(b)** kite (in general) | Kite need not be parallel |
| 6 | **(c)** 13 cm | Diagonals bisect at right angles; use Pythagoras |
| 7 | **(a)** Yes | Difference of sums of alternate digits |
| 8 | **(b)** associative property | Grouping does not matter |
| 9 | **(c)** 4 : 5 | Divide both by HCF 7 |
| 10 | **(a)** 0.75 kg | Unit conversion |
| 11 | **(b)** 225 | 45 × 5 |
| 12 | **(d)** 25% | Profit or loss on CP |
| 13 | **(a)** (3, 4, 5) | m=2, n=1 |
| 14 | **(a)** No | $9^2 + 12^2 \neq 16^2$ |
| 15 | **(b)** 100 km | Multiply by the scale |
| 16 | **(a)** 6 | Count the edges |
| 17 | **(c)** 21 | Mean of the two middle values |
| 18 | **(c)** 5 | $(2n + 10)/2 - n = 5$ |
| 19 | **(d)** 50 cm² | $\frac{1}{2}d^2$ |
| 20 | **(a)** 120 cm² | $l \times b$ |

### SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $30000 < 200000$, so $2 \times 10^5$ is larger (2). | 2 |
| 22 | It holds a place when a position is empty, distinguishing 305 from 35 (2). | 2 |
| 23 | Sum $10 + x$ must be 18, so $x = 8$ (2). | 2 |
| 24 | $120 : 40 = 3 : 1$ (2). | 2 |
| 25 | 29 m (2). | 2 |
| 26 | 75 (2). | 2 |
| 27 | Mean $x + 2 = 10$; $x = 8$ (2). | 2 |
| 28 | Other diagonal 24 (1); area 120 cm² (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 8 × 3 = 24

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 29 | Volume $= 13.824$ m³ (1); edge $= \sqrt[3]{13.824} = 2.4$ m (2). | 1+2 |
| 30 | Angle sum $= 360^\circ$ (1); $C = 360 - 90 - 90 - 70$ (1); $110^\circ$ (1). | 1+1+1 |
| 31 | $x^2 + 3x - 10$ (2); 44 (1). | 2+1 |
| 32 | SI $=$ ₹1800 (2); amount $=$ ₹10800 (1). | 2+1 |
| 33 | One base and seven triangles (3). | 3 |
| 34 | $78 + 87 = 165 = 11 \times 15$ (1.5); $11(a + b)$ (1.5). | 1½+1½ |
| 35 | Diagonal 15 m (1.5); saved $21 - 15 = 6$ m (1.5). | 1½+1½ |
| | **OR:** $4 + 9 \neq 16$ (1.5); $81 + 144 = 225$ (1.5). | 1½+1½ |

| | | |
|---|---|---|
| **36** | Room 48 m², carpet 24 m² (2); uncovered 24 m² (1). | 2+1 |
| | **OR:** Triangle 30 cm² (1); parallelogram 60 cm² (1); ratio 1 : 2 (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**4 × 5 = 20**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **37** | (i) Compact, easy to compare and calculate (2). (ii) $9.1 \times 10^{-31}$ kg (3). | 2+3 |
| | **OR:** (i) $a^{m-m} = a^0 = 1$ (3). (ii) $9 \times \frac{1}{9} = 1$ (2). | 3+2 |
| **38** | (i) 37.5% and 62.5% (2). (ii) 28 L (3). | 2+3 |
| | **OR:** (i) $P \times 1.1664 = 5832$, $P = 5000$ (3). (ii) Interest added to principal each period (2). | 3+2 |
| **39** | (i) $x = 8$ (2). (ii) 7.2 (3). | 2+3 |
| | **OR:** (i) Mean 8; deviations $-4, -2, 0, 2, 4$ sum 0 (3). (ii) Left and right deviations balance (2). | 3+2 |
| **40** | (i) Both equal base × common height (3). (ii) Base 10, height 6 gives 60 for both (2). | 3+2 |
| | **OR:** (i) 360 cm² (2). (ii) Foot distance 8 cm; area $\frac{1}{2} \times 8 \times 15 = 60$ cm² (3). | 2+3 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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